SCOTT TRC-1 AIR CART — QUICK REFERENCE

Model 200070-01 | 6-Outlet Cart, Hansen Fittings | Source: Scott Safety O&M Instructions P/N 595269-01 Rev. A (6/12) — this sheet is a summary; refer to the full manual for complete procedures and warnings.

WHAT IT DOES

Wheeled cascade cart supplying breathing air to respirators (SCBA extended-duration hoseline, Type C airline) from two onboard cylinders and/or a remote high-pressure line. Two separate manifolds: **Respirator Supply** (0–100 psi, breathing air only) and **Auxiliary Supply** (tools or backup). Supports up to 8 respirator users depending on configuration.

AIR SOURCE SELECTOR

Setting	Effect
Common Air Source	Both manifolds draw from whichever source is opened (cylinders or remote line).
Isolated Air Source	Respirator Supply = cylinders ONLY. Auxiliary Supply = remote line ONLY. Always use this setting when running pneumatic tools.

SETUP — CYLINDERS ONLY

1. Both outlet pressure regulators turned fully counterclockwise (zero) BEFORE opening any valve.
2. Confirm both cylinders full, cylinder valves closed, vent valves closed.
3. Slowly open one cylinder valve — alarm chirps briefly until pressure passes 1000 psi, then resets.
4. Turn Respirator Outlet Regulator and Auxiliary Regulator up to 60–100 psi with nothing connected; check for leaks/hiss.
5. Connect respirator hoses — confirm a firm "click" on each quick-disconnect; tug-test.

LOW-PRESSURE ALARMS

Whistle = cylinder supply low. **Bell** = remote high-pressure line low. Both trip at approx. **500 psi** and CANNOT be silenced — they sound continuously until pressure is restored above 1000 psi or the cylinder empties.
On alarm: open the reserve cylinder → close & vent the depleted one → swap in a fresh, fully-charged cylinder immediately.

HOT-SWAPPING A CYLINDER MID-USE

Users stay on air the whole time — always open the fresh cylinder BEFORE closing the depleted one.

1. When the whistle sounds (or gauge reads low), open the valve on the UNUSED cylinder.
2. Wait for inlet pressure to climb back above 1000 psi — the alarm resets itself automatically.
3. Now close the valve on the depleted cylinder (push in, turn clockwise).
4. Crack open that cylinder's vent valve to bleed the trapped line pressure.
5. Once vented, disconnect the high-pressure coupling (hand-tight only — no wrench), loosen the retention strap, and pull the empty cylinder.
6. Strap in a fresh, fully-charged cylinder, valve outlet oriented toward the coupling; confirm the gasket is present and undamaged; hand-tighten the coupling.

CONNECTING AN EXTERNAL SOURCE (AIR TRUCK / CASCADE)

The cart's High Pressure Inlet accepts a remote high-pressure hose line up to 5500 psi.

- The supply hose must have a shut-off valve at the outlet end.
- The external source must hold the respirator's minimum required outlet pressure all the way down to approx. 500 psi — verify this in a clean air area before relying on it.
- A check valve isolates the remote inlet from the onboard cylinders — truck air CANNOT be used to recharge the cart's two cylinders, it only feeds the manifolds directly while connected.
- Common Air Source: truck air feeds both Respirator and Auxiliary manifolds; cylinders sit as backup.
- Isolated Air Source: truck air is dedicated to Auxiliary only (e.g., running tools) while cylinders exclusively supply the Respirator manifold — use this setting any time the truck may also run pneumatic tools, so tool air never touches the breathing air path.

- Standard practice: keep two fully charged cylinders loaded as backup any time the cart is running off an external/truck source.

SHUTDOWN SEQUENCE

1. Confirm all respirator users are in a safe (non-hazard) area.
2. Users doff respirators and disconnect hoses from the cart.
3. Fully close both cylinder valves (push in, turn clockwise).
4. Close and disconnect remote high-pressure line, if used.
5. Slowly open both vent valves to bleed remaining system pressure.
6. Vent trapped air: partially engage a male quick-disconnect into one Respirator outlet and one Auxiliary outlet.
7. Turn both outlet pressure regulators fully counterclockwise (zero).
8. Move cart to safe area; clean, inspect, and re-rack per Maintenance section.

CRITICAL SAFETY RULES

■ NEVER connect tools to Respirator Outlets — lubricant contamination risk. Tools go on Auxiliary Outlets only.

■ Once Auxiliary Outlets have supplied tools, they are NOT to be used for respirators again until air quality is verified clean (Grade D, CGA G-7.1 or better).

■ NOT rated for IDLH atmospheres unless the respirator itself carries an emergency escape/backup air supply.

■ A trained attendant with two-way comms MUST monitor the cart at all times while it's supplying respirators.

■ Hand-tighten high-pressure couplings ONLY — never use a wrench (damages the coupling gasket).

■ Air tools can rapidly deplete cylinder air — always use Isolated Air Source when tools are running so breathing air is protected.

■ Use only clean breathing air (CGA G-7.1 Grade D+, dew point ≤ -65°F) — never oxygen or shop air.

HOSE LIMITS (Table 1, summarized)

Maintain 60–100 psi while flowing ≥200 LPM per user. Approved total hose length: Hansen HK coupling (26369 series) — up to 150 ft / 1 segment. Foster, standard Hansen, Schrader, and CEJN coupling series — up to 300 ft / up to 3 segments. Always test couplings match your respirator equipment before relying on them in the field.

MAINTENANCE AFTER EACH USE

- Damp-sponge dirt from cart, cylinders, and fittings — don't obscure warning labels.
- Perform Regular Operational Inspection (alarm test, coupling check, leak check).
- Install two fully charged cylinders before returning to service.
- Store in a cool, dry area.
- Tag out and remove from service immediately if any damage, leak, or alarm failure is found.

This is a condensed field/training reference, not a substitute for the full Scott Safety O&M; manual (P/N 595269-01). Full manual: fedfiretrainingsd.org — search "Scott TRC-1 air cart manual." Scott Safety: 1-800-247-7257 · www.scottsafety.com